

Lecture 36: Nov 19

Last time

- Joint pmf

Today

- Course Evaluations (1/42)
- Joint cdf
- Joint pdf

Bivariate cdfs Whether they are discrete or continuous or some combination of the two, we can always define the *joint cdf*. For $n = 2$, the *bivariate cumulative distribution function* is

$$F_{X,Y}(x, y) = \Pr\{X \leq x, Y \leq y\}$$

Properties:

- $F_{X,Y}(x, y) \geq 0$
- $F_{X,Y}(\infty, \infty) = 1$
- $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$
- $F_{X,Y}(-\infty, -\infty) = 0$
- F is non-decreasing and right-continuous in each variable separately.

Joint probabilities All joint probability statements about X and Y can be answered in terms of their joint cdf:

$$\begin{aligned} \Pr(x_1 < X \leq x_2, y_1 < Y \leq y_2) = \\ F_{X,Y}(x_2, y_2) + F_{X,Y}(x_1, y_1) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) \end{aligned}$$

Example

$$\Pr(X > x, Y > y) = 1 - F_X(x) - F_Y(y) + F_{X,Y}(x, y)$$

Note: To ensure that a bivariate function $F(x, y)$ is a proper cdf, it must satisfy all the properties mentioned above and the rectangular property above.

Marginal distributions From $F_{X,Y}$, we can derive the univariate distribution functions for X and Y . These are generally called *marginal distributions*.

$$\begin{aligned} F_X(x) &= \Pr\{X \leq x\} = \Pr\{X \leq x, Y \leq \infty\} = F_{X,Y}(x, \infty) \\ F_Y(y) &= \Pr\{Y \leq y\} = \Pr\{X < \infty, Y \leq y\} = F_{X,Y}(\infty, y) \end{aligned}$$

Note: Although we can obtain $F_X(x)$ and $F_Y(y)$ from the joint cdf, we cannot do the reverse.

Continuous Bivariate RVs The random variables X and Y are said to be *jointly continuous* if there exists a function $f_{X,Y}(x, y)$, such that for any Borel set B of 2-tuples in $\mathbb{R}^2$,

$$\Pr\{(X, Y) \in B\} = \int \int_{(x,y) \in B} f_{X,Y}(x, y) dx dy.$$

The function $f_{X,Y}(x, y)$ is called the *joint probability density function* for X and Y . It follows in this case that

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(s, t) dt ds,$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y}$$

Properties of the bivariate pdf

- $f_{X,Y}(x, y) \geq 0$
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$
- $f_{X,Y}(x, y)$ is not a probability, but can be thought of as a relative probability of (X, Y) falling into a small rectangle located at (x, y) :

$$\Pr\{x < X \leq x + dx, y < Y \leq y + dy\} \approx f(x, y) dx dy$$

- The *marginal probability density functions* for X and Y can be obtained as

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Example 1

$$F_{X,Y}(x, y) = xy \quad 0 < x \leq 1, 0 < y \leq 1$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} =$$

$$f_X(x) =$$

$$f_Y(y) =$$

Example 2

$$F_{X,Y}(x, y) = x - x \log \frac{x}{y} \quad 0 < x \leq y \leq 1$$

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \partial y} =$$

$$f_X(x) =$$

$$f_Y(y) =$$

Note: Once we have $f_X(x)$ and $f_Y(y)$, we can obtain $F_X(x)$ and $F_Y(y)$ directly. Double check: $F_X(x) = F_{X,Y}(x, \infty)$.

Conditional Distributions

Conditional Distributions - Discrete Recall if A and B are two events, the probability of A conditional on B is:

$$\Pr(A|B) = \frac{\Pr(A, B)}{\Pr(B)}$$

Defining the events $A = \{Y = y\}$ and $B = \{X = x\}$, it follows that

$$\begin{aligned}\Pr\{Y = y|X = x\} &= \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} \\ &= \frac{f_{X,Y}(x, y)}{f_X(x)} \\ &= f_{Y|X}(y|x)\end{aligned}$$

This is called the *conditional probability mass function* of Y given X .

Example: Discrete Back to the fair coin example. From the joint pmf of X and Y , we can derive all the conditional pmfs:

		Y			
		0	1	2	3
X	0	1/8	1/4	1/8	0
	1	0	1/8	1/4	1/8

Conditional Distribution - Continuous If $F(x, y)$ is absolutely continuous, we define the conditional density of X given Y as:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \text{ if } f_Y(y) > 0$$

Example 1

$$\begin{aligned}F_{XY}(x, y) &= xy && 0 < x < 1, \quad 0 < y < 1 \\ f_{XY}(x, y) &= 1 && 0 < x < 1, \quad 0 < y < 1 \\ f_X(x) &= 1 && 0 < x < 1 \\ f_Y(y) &= 1 && 0 < y < 1 \\ f_{X|Y}(x|y) &= \frac{f_{XY}(x, y)}{f_Y(y)} = 1 && 0 < x < 1 \quad (0 < y < 1) \\ f_{Y|X}(y|x) &= \frac{f_{XY}(x, y)}{f_X(x)} = 1 && 0 < y < 1 \quad (0 < x < 1)\end{aligned}$$

Note: Here we get that the conditional densities are the same as the marginals. This means X and Y are independent.

Example 2

$$\begin{aligned}
F_{XY}(x, y) &= x - x \log \frac{x}{y} & 0 < x \leq y \leq 1 \\
f_{XY}(x, y) &= 1/y & 0 < x \leq y \leq 1 \\
f_X(x) &= -\log x & 0 < x \leq 1 \\
f_Y(y) &= 1 & 0 < y \leq 1 \\
f_{X|Y}(x|y) &= \frac{f_{XY}(x, y)}{f_Y(y)} = 1/y & 0 < x \leq y \quad (0 < y \leq 1) \\
f_{Y|X}(y|x) &= \frac{f_{XY}(x, y)}{f_X(x)} = -\frac{1}{y \log x} & x \leq y \leq 1 \quad (0 < x \leq 1)
\end{aligned}$$

- Y is marginally uniform, but not conditionally uniform.
- X is conditionally uniform, but not marginally uniform.

Independent Random Variables

Independence The random variable X and Y are said to be *independent* if for any two Borel sets A and B ,

$$\Pr(X \in A, Y \in B) = \Pr(X \in A) \Pr(Y \in B)$$

All events defined in terms of X are independent of all events defined in terms of Y .

Using the Kolmogorov axioms of probability, it can be shown that X and Y are independent if and only if $\forall(x, y)$ (except possibly for sets of probability 0)

$$F_{X,Y}(x, y) = F_X(x)F_Y(y)$$

or in terms of pmfs (discrete) and pdfs (continuous)

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

Checking independence

- A necessary condition for independence of X and Y is that their joint pdf/pmf has positive probability on a rectangular domain.
- If the domain is rectangular, one can try to write the joint pdf/pmf as a product of functions of x and y only.

Example Two points are selected randomly on a line of length a so as to be on opposite sides of the mid-point of the line. Find the probability that the distance between them is less than $a/3$.

Solution:

Let X be the coordinate of a point selected randomly in $[0, a/2]$ and Y be the coordinate of a point selected randomly in $[a/2, a]$. Assume X and Y are independent and uniform over its interval. The joint density is

$$f_{X,Y}(x, y) = 4/a^2, \quad 0 \leq x \leq a/2, a/2 \leq y \leq a$$

Therefore, the solution is

$$\Pr(Y - X < a/3) =$$

Example: Buffon's Needle A table is ruled with lines distance 1 unit apart. A needle of length $L \leq 1$ is thrown randomly on the table. What is the probability that the needle intersects a line?

Solution:

Define two random variables:

- X : distance from low end of the needle to the nearest line above
- θ : angle from the vertical to the needle.

By "random", we assume X and θ are independent, and

$$X \sim U(0, 1) \quad \text{and} \quad \theta \sim U[-\pi/2, \pi/2].$$

This means that

$$f_{X,\theta}(x, \theta) = 1/\pi, \quad 0 \leq x \leq 1, -\pi/2 \leq \theta \leq \pi/2$$

For the needle to intersect a line, we need $X < L \cos(\theta)$.

Expectations of Independent RVs (Theorem 4.2.10) Let X and Y be independent rvs.

- For any $A \subset \mathbb{R}$ and $B \subset \mathbb{R}$,

$$\Pr(X \in A, Y \in B) = \Pr(X \in A) \Pr(Y \in B)$$

i.e., the events $\{X \in A\}$ and $\{Y \in B\}$ are independent.

- Let $g(x)$ be a function only of x and $h(y)$ be a function only of y . Then

$$E[g(X)h(Y)] = [Eg(X)][Eh(Y)]$$

Proof:

$$\begin{aligned} E[g(X)h(Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y)f_{XY}(x, y)dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y)f_X(x)f_Y(y)dx dy \\ &= \left(\int_{-\infty}^{\infty} g(x)f_X(x)dx \right) \left(\int_{-\infty}^{\infty} h(y)f_Y(y)dy \right) \\ &= [Eg(X)][Eh(Y)] \end{aligned}$$

Example X, Y are independent

$$\begin{aligned} E(X^2Y^3) &= (EX^2)(EY^3) \\ E(Y^2Y^3) &\neq (EY^2)(EY^3) \end{aligned}$$